SEQUENCE 8

Algebra, equations and inequalities

4–20 January 2027

8.1 Brackets and distributivity

Method · Removing brackets. After a $+$ sign, removing the brackets keeps the signs of the terms. After a $-$ sign, every term inside changes sign. This rule applies to a sum or a difference of terms.

Example 1. Remove the brackets and simplify: $A = x + (2 + 3x)$; $B = x - (2 + 3x)$; $C = -2x + (-4 + 5x)$; $D = -2x - (-4 + 5x)$.

Key point · Distributivity. For all real numbers a, b, c, d:

$$b(c + d) = bc + bd, \qquad (a + b)(c + d) = ac + ad + bc + bd.$$

Example 2. Expand and simplify:

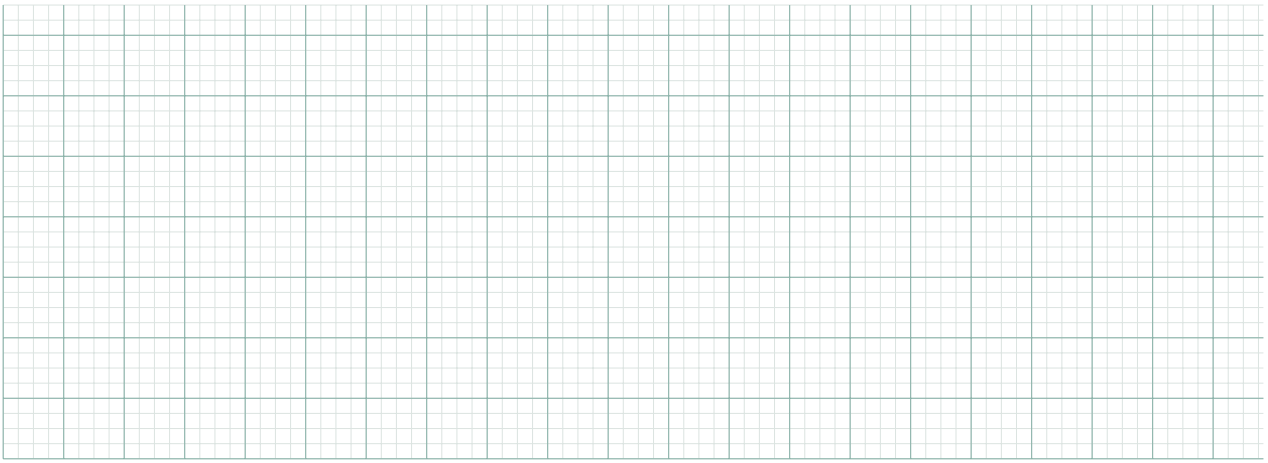
$$\begin{aligned} A &= -4(8x + 7) + (-8x + 5)(5x + 9), \\ B &= (6x + 9)(-x + 7) + (3x + 5)(-9x + 6), \\ C &= -9(-x - 7) - (-x + 3)(-8x - 8), \\ D &= -5(-6x - 7) + (x - 7)(5x + 8). \end{aligned}$$

8.2 Factorising with a common factor

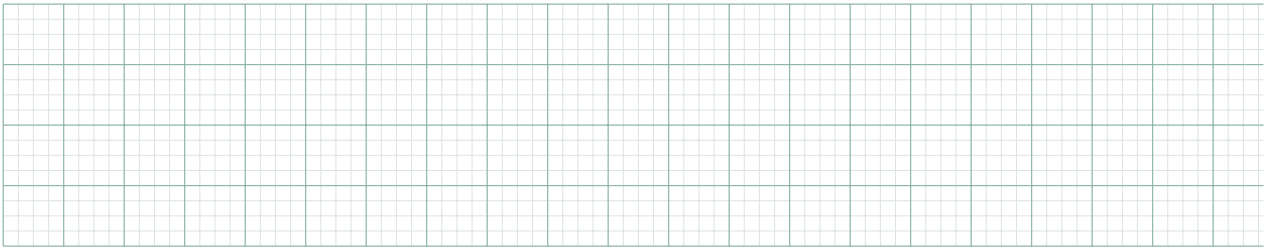
Key point · Property. $ka + kb = k(a + b)$ and $ka - kb = k(a - b)$. Factorising turns a sum into a product; expanding does the reverse.

Example 3. Factorise:

$$A = 15x^2 - 21x,$$
$$B = 3(2 + 3x) - (5 + 2x)(2 + 3x),$$
$$C = (4x - 1)(x + 6) + (4x - 1),$$
$$D = (1 - 6x)^2 - (1 - 6x)(2 + 5x).$$



Example 4. Choose a useful form of $f(x) = (x - 2)(x + 3)$ to calculate $f(2)$, $f(0)$ and find its zeros.



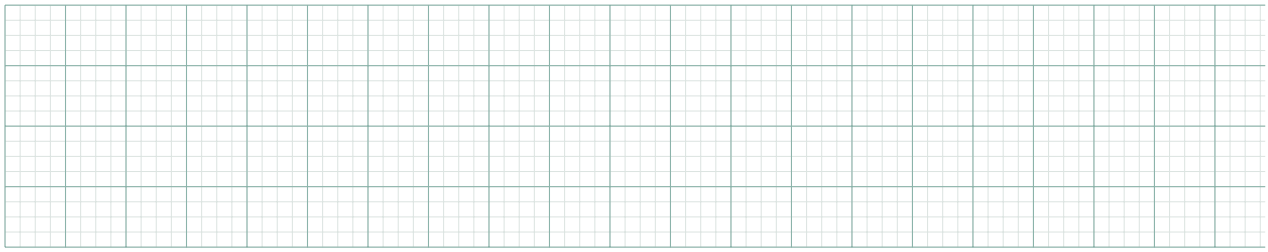
8.3 Special identities: expanding

Key point · Identities.

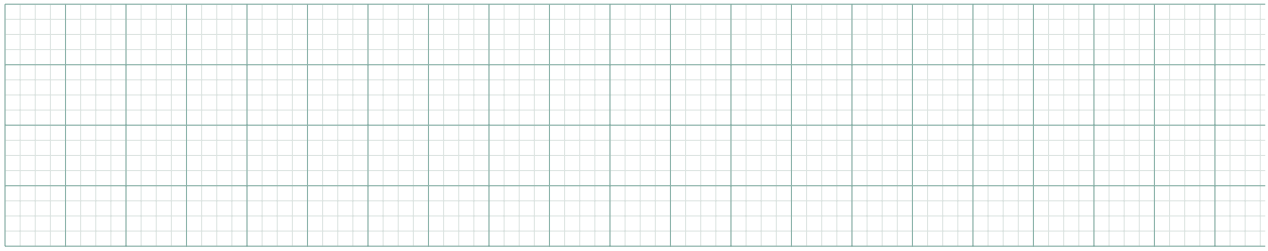
$$(a+b)^2 = a^2 + 2ab + b^2, \qquad (a-b)^2 = a^2 - 2ab + b^2,$$

$$(a+b)(a-b) = a^2 - b^2.$$

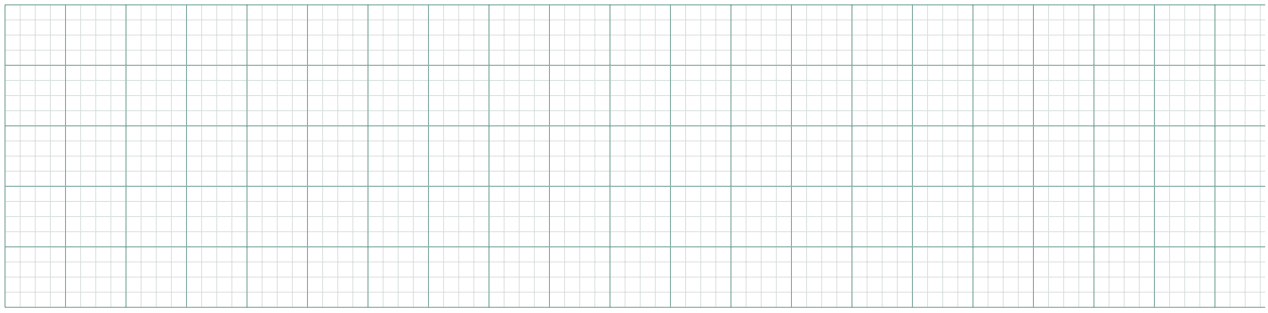
Proof to write.



Example 5. Expand $(x+3)^2$, $(3x-4)^2$, $(x-3)(x+3)$, $(2x+3)(2x-3)$, $(4-3x)(3x+4)$.



Example 6. Expand and simplify $A = (2x-3)^2 + (x+5)(3-x)$ and $B = 2(x+3) + (2x+3)(2x-3)$.



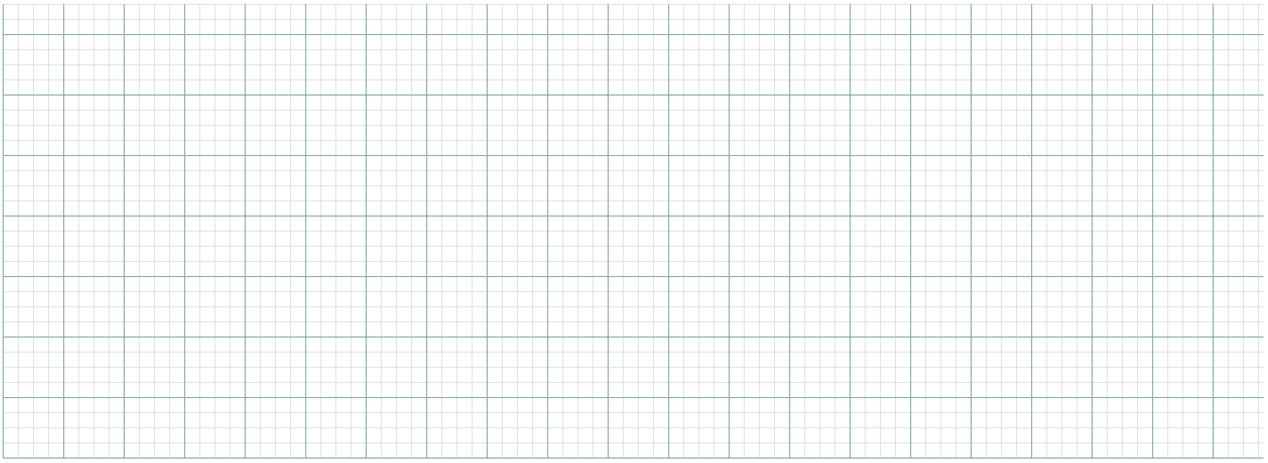
8.4 Special identities: factorising

Method. Spot the two squares; for three-term forms, check the double product $2ab$. For a difference of two squares, use $a^2 - b^2 = (a - b)(a + b)$.

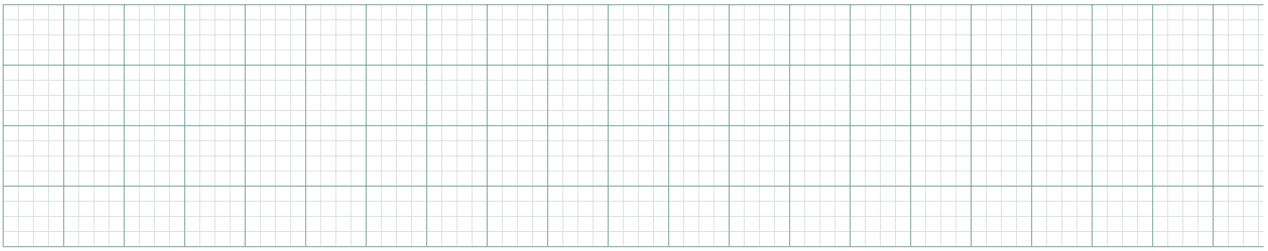
Example 7. Factorise:

$$A = x^2 - 2x + 1, \quad B = 25 + 16x^2 - 40x, \quad C = 4x^2 + 12x + 9,$$

$$D = 1 - 49x^2, \quad E = 9x^2 - 4, \quad F = (2x + 3)^2 - 64.$$



Example 8 · Checking an identity. Are the equalities $(a + b)^2 = a^2 + b^2$ and $a^2 + b^2 = (a - b)(a + b)$ true for all real numbers?



Fluency · 5–10 min. Calculate 101^2 mentally; give a direction vector of $2x + 3y = 1$; determine $[-1, 3] \cap [2, 5]$.

8.5 Integer powers

Key point · Definition and rules. For $n \geq 1$, a^n is the product of n factors equal to a. For $a \neq 0$, $a^0 = 1$ and $a^{-n} = 1/a^n$. In this course, 0^0 is not defined. For $a, b \neq 0$ and integers m, n:

$$a^m a^n = a^{m+n}, \quad \frac{a^m}{a^n} = a^{m-n}, \quad (a^m)^n = a^{mn},$$

$$(ab)^n = a^n b^n, \qquad \left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}.$$

For strictly positive exponents, some bases may be zero if the expressions remain defined.

Example 9. Simplify $A = 2^4 \times 2^3$, $B = (-5)^{12}/(-5)^3$, $C = (7^3)^6 \times 7^{-5}$, $D = 7^{24} \times 3^2 \times 2^3$. Also calculate $(-3)^5$, 5^1 and 99999^0.

Watch out. The definition as a repeated product does not hold for a negative exponent. $-3^2 = -9$, whereas $(-3)^2 = 9$.

Example 10. Solve $2^n > 100$ mentally for the smallest natural number n. Write a program that finds it.

8.6 Fractions and fractional expressions

Key point · Rules and conditions. For nonzero denominators:

$$\frac{a}{c} \pm \frac{b}{c} = \frac{a \pm b}{c}, \quad \frac{a}{c} \pm \frac{b}{d} = \frac{ad \pm bc}{cd}, \quad \frac{a}{c} \times \frac{b}{d} = \frac{ab}{cd}.$$

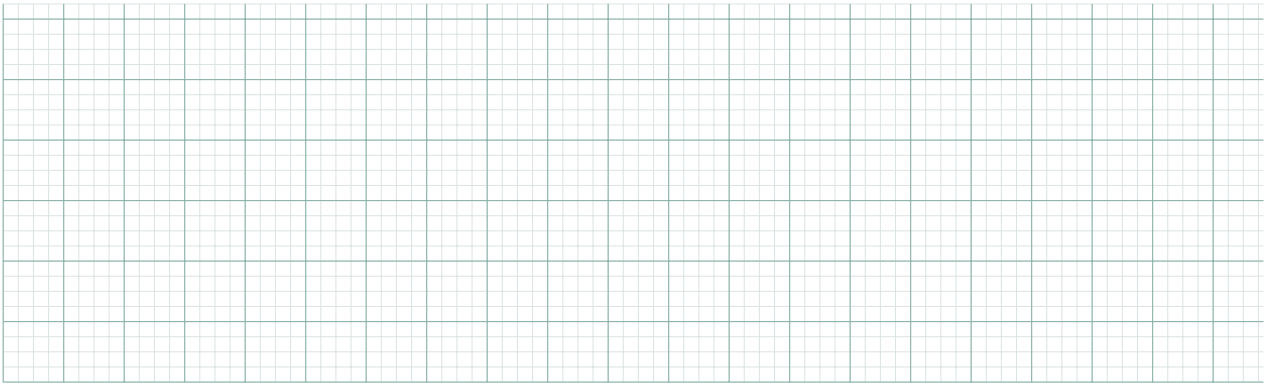
A fraction c/d has an inverse only if $c \neq 0$ and $d \neq 0$. Then

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c} = \frac{ad}{bc} \quad (b, c, d \neq 0).$$

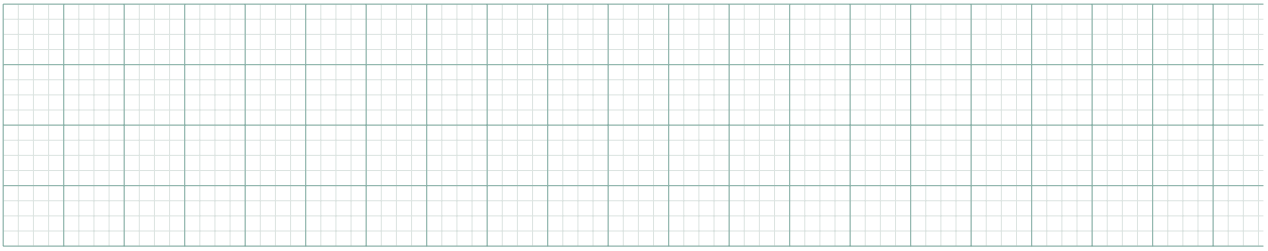
Example 11. Calculate and reduce to lowest terms:

$$A = \left(\frac{2}{3} + \frac{1}{6}\right) \times \frac{4}{5}, \quad B = \frac{1}{2} + \frac{3}{4} \times \frac{5}{6}, \quad C = \frac{4}{5} \div \frac{2}{3} \times \frac{7}{10},$$

$$D = \frac{\left(\frac{2}{5} + \frac{1}{3}\right) \times \frac{3}{4}}{\frac{1}{2} - \frac{1}{6}}.$$



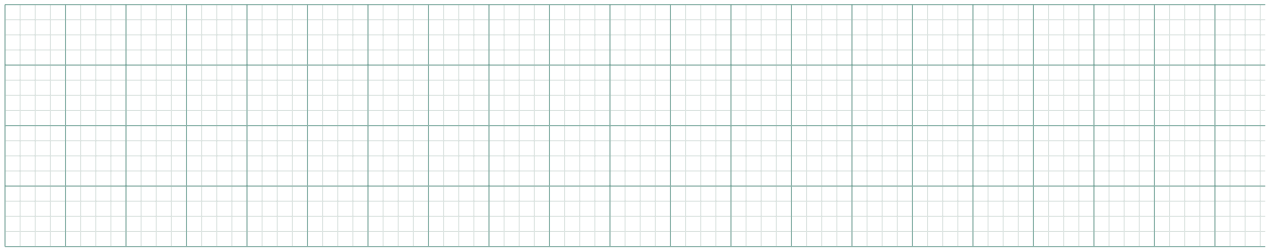
Example 12. Simplify $\dfrac{x^2 - 1}{x - 1}$, stating the domain. Can we then take $x = 1$?



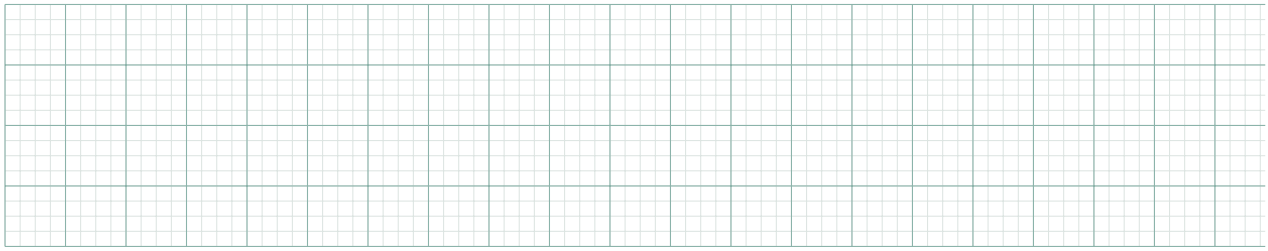
8.7 Square roots and validity conditions

Key point · Definition and properties. For $a \geq 0$, $\sqrt{a}$ is the unique positive or zero number whose square is a. Thus $(\sqrt{a})^2 = a$. For every $x \in \mathbb{R}$, $\sqrt{x^2} = |x|$. For $a, b \geq 0$, $\sqrt{ab} = \sqrt{a}\sqrt{b}$. If $b > 0$, $\sqrt{a/b} = \sqrt{a}/\sqrt{b}$.

Proof to write.

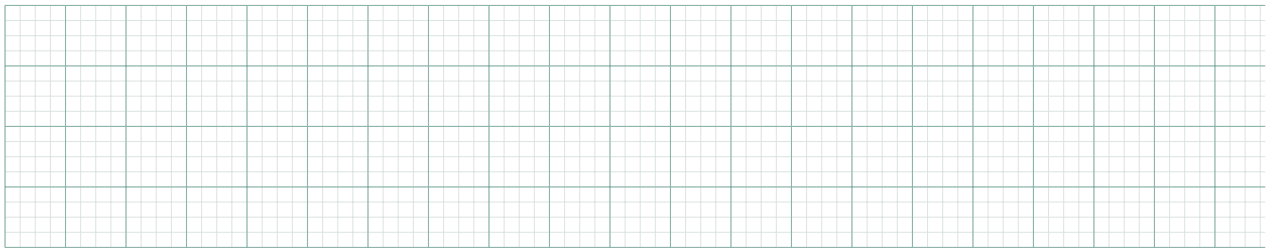


Example 13. Simplify $\sqrt{18}$; $\sqrt{2}\sqrt{8}$; $\sqrt{7/25}$; $\sqrt{27/75}$; $\sqrt{(-5)^2}$.



Watch out. There is no general rule $\sqrt{a+b} = \sqrt{a} + \sqrt{b}$. With $a = 4, b = 9$, we would wrongly have $\sqrt{13} = 5$. Saying these expressions are always different would also be wrong, for example if $a = 0$.

Example 14. Simplify $2\sqrt{18} - \sqrt{50} + \sqrt{8}$. Bound $\sqrt{20}$ between two consecutive integers.



8.8 First-degree equations

Key point · Definition. To solve an equation is to find all the values of the unknown that satisfy the equality. Two equations are equivalent if they have the same solutions. Adding the same expression or multiplying by the same nonzero number preserves equivalence.

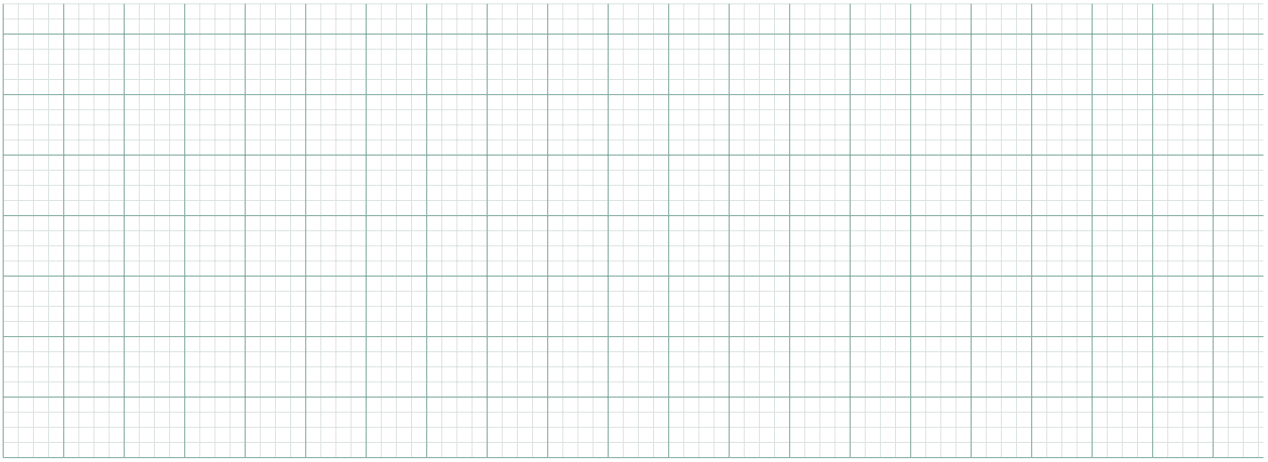
Method. Expand if necessary, collect the terms in x, then isolate x. If its coefficient is zero, examine the remaining equality: it may be always true or impossible. Check in the original equation.

Example 15. Solve:

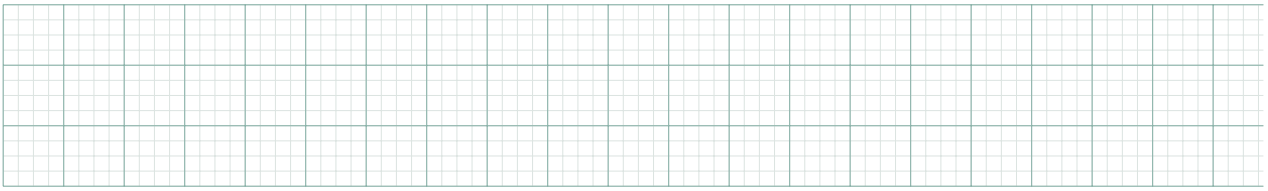
$a)\ 5x-3=2x+9,$ $b)\ 3(2x-1)=2(x+4)+7,$

$c)\ -2(x+3)=4x-(x-1),$ $d)\ \dfrac{x+2}{3}=x-1,$

$e)\ \dfrac{-5x+5}{8}=-\dfrac{3}{4},$ $f)\ \dfrac{-3x-1}{9}=-\dfrac{7}{4}.$



Example 16. Solve $2(x+1)=2x+2$, then $2(x+1)=2x+3$.



8.9 Zero-product property and equations with a quotient

Key point · Properties. $AB = 0 \iff A = 0$ or $B = 0$. For $B \neq 0$, $A/B = 0 \iff A = 0$. To solve $A(x)/B(x) = k$, first require $B(x) \neq 0$, then solve $A(x) = kB(x)$ and discard the forbidden values.

Example 17. Solve $(4x+6)(3-7x) = 0$; $3x(-7x+5) = 0$; $\dfrac{2x+1}{3x-5} = 0$.

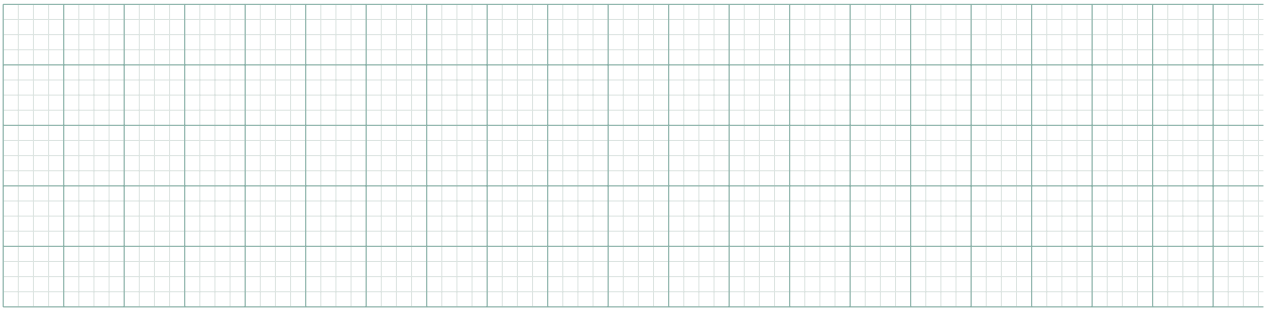
Example 18. Factorise then solve $(1-6x)^2 - (1-6x)(2+5x) = 0$ and $(4x-1)(x+6) + (4x-1) = 0$.

Example 19. Solve $\dfrac{x+1}{x-2} = 3$, then $\dfrac{x^2-4}{x-2} = 4$.

8.10 Equations with a square; isolating a variable

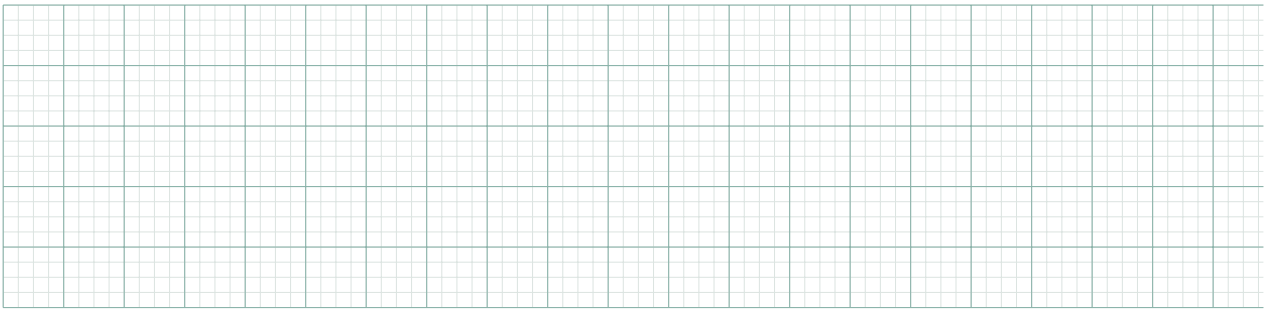
Key point · Reminder. In $\mathbb{R}$, $x^2 = a$ has two solutions $\pm\sqrt{a}$ if $a > 0$, one solution 0 if $a = 0$, and none if $a < 0$.

Example 20. Solve $x^2 = 0$; $x^2 = -8$; $x^2 = 7$; $25x^2 = 81$; $(x+2)^2 = 9$.



Method · Isolating a variable in a formula. Describe the operations performed on both sides; state the conditions that allow dividing or taking a square root.

Example 21. Isolate h in $V = \pi r^2 h$ with $r > 0$; isolate v in $E = \frac{1}{2}mv^2$ with $m > 0$, $E \geq 0$ and $v \geq 0$; isolate x in $y = ax + b$ with $a \neq 0$.

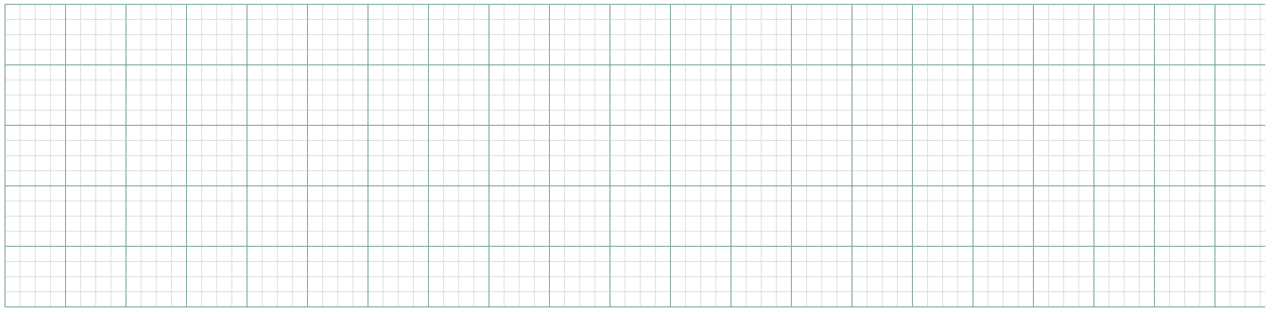


Fluency · 5–10 min. Give the domain of $1/(x+3)$; calculate $\sqrt{49}$; the slope between $(0,2)$ and $(3,8)$.

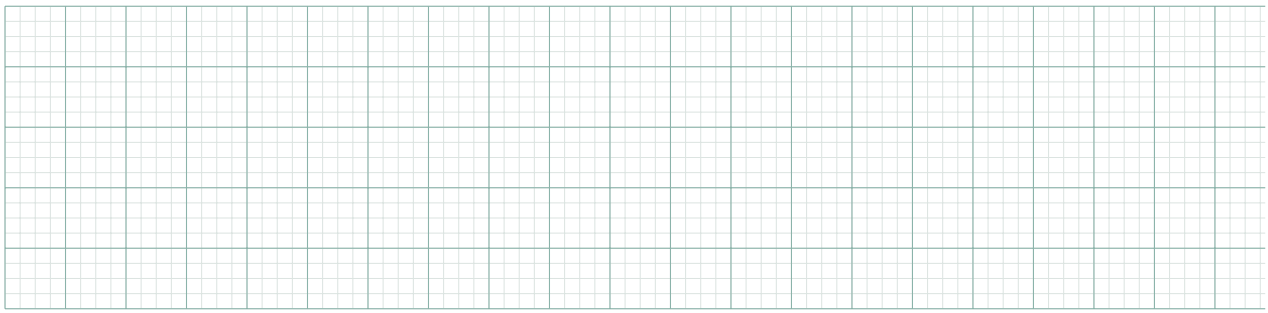
8.11 Inequalities and comparison

Key point · Rules. Inequalities of the same direction can be added: $a \leq b$, $c \leq d \Rightarrow a + c \leq b + d$. Adding the same real number preserves the direction. Multiplying or dividing by a strictly positive real number preserves the direction; by a strictly negative one, it reverses it. Comparing A and B amounts to studying the sign of $A - B$. If $A, B > 0$, one can also compare A/B to 1.

Example 22. 1. If $x \leq -4$, what can be said about $3x - 4$? 2. If $x > -6$, what can be said about $2x - 3$? 3. If $7 \leq x \leq 16$ and $3 \leq y \leq 12$, bound $8x + 7y$. 4. If $-10 \leq x \leq -6$, bound $2x - 6$.



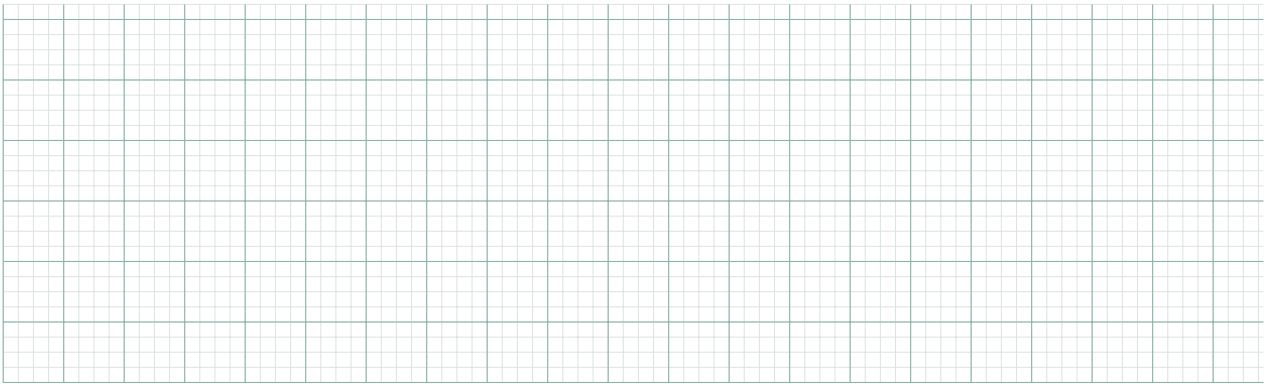
Example 23. Solve $3x + 1 \geq x - 7$; $2x - 11 \leq 4x - 5$; $6x + 19 \leq 2x - 3$.



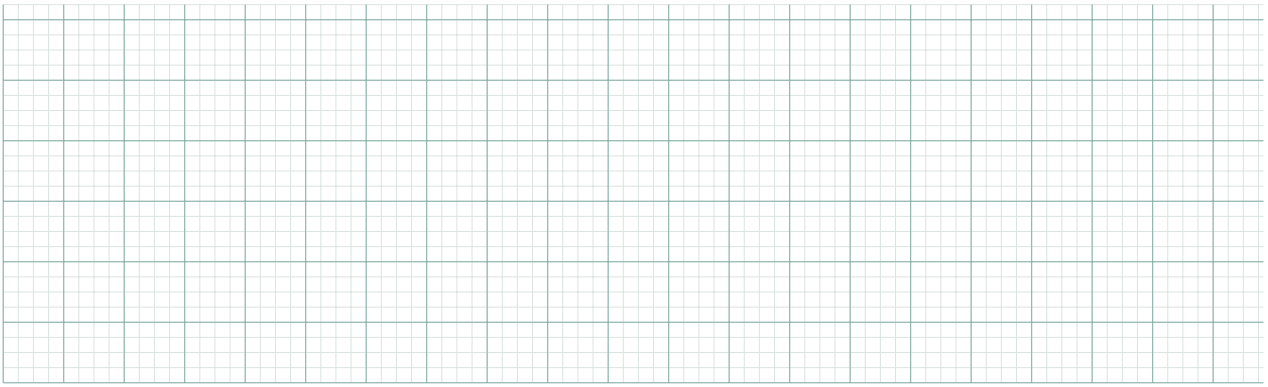
8.12 Sign of an affine expression and of a product

Key point · Property. For $a \neq 0$, $ax + b$ vanishes at $x_0 = -b/a$; its sign to the right is that of a, to the left the opposite. If $a = 0$, the sign is that of the constant b. For a product, multiply the signs of the factors.

Example 24. Draw the sign tables of $3 - x$, $2x - 6$, then $P(x) = (2x - 6)(x + 1)$.



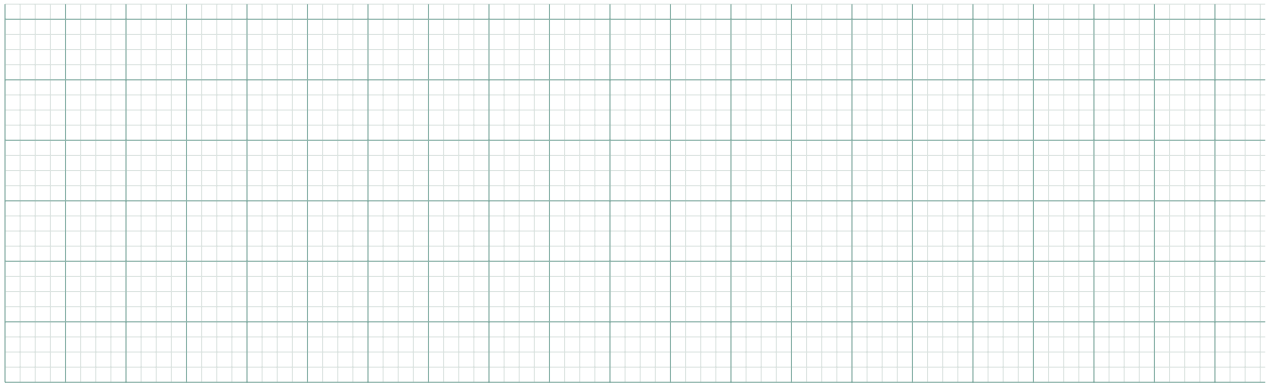
Example 25. Solve $(7 + 3x)(2x + 6) > 0$.



8.13 Sign of a quotient and algorithmic synthesis

Method. Find the forbidden values, then the zeros of the numerator. Order all the critical values and apply the sign rule. On the quotient row, a double bar indicates a forbidden value.

Example 26. Draw the sign table of $Q(x) = \dfrac{-4x+6}{x-3}$, then solve $\dfrac{2x-1}{-4x+3} \leq 0$.



```
def premiere_puissance(seuil):
    # seuil reel >= 1
    n, p = 0, 1
    while p <= seuil:
        n = n+1
        p = 2*p
    return n, p
```

Example 27. What does `premiere_puissance(100)` return? What relation between `p` and `n` stays true at each step? How can the sign of a number then be tested in Python?

